Technique Packet No. 4

Count the Opposite — when the thing you want is hard to count, count the rest.

Name: ______________________ Date: ______________________

A clubhouse door has a secret code: two symbols, and each one is a star, a moon, or a sun. (You may use a symbol twice.) The door opens only if the code has **at least one star**. How many codes open the door?

You could hunt for every code with a star in it and hope you catch them all. Or you could notice that the *opposite* is much easier to count: the codes with **no star at all**. There are 3 × 3 = 9 codes in total, and the ones with no star use only moon or sun — 2 × 2 = 4 of them. So the codes with at least one star are simply 9 – 4 = 5. This is the whole trick: **when what you want is hard to count head-on, count everything, then subtract the part you don't want.**

EVERYTHING (the total)

what you WANT = EVERYTHING – the part you DON'T want

Count the opposite when it is the easier thing to count.

Count the whole bar, shade off the part you don't want, and what is left is your answer — no hunting required.

THE TECHNIQUE — Count the Opposite.
When counting what you want directly would mean lots of separate cases:

1. Count *everything* — the whole total, ignoring the condition.
2. Count the *opposite* — just the ones you don't want. (This is usually the easy part.)
3. Subtract: total – opposite = what you want.

When to reach for it: watch for "at least one," "not," "none," "no —", or "not all the same." If counting it directly splits into many cases but the opposite is one tidy group, count the opposite and subtract.

Watch how it works

WORKED EXAMPLE 1

The clubhouse code is two symbols, each a star, moon, or sun (repeats allowed). How many codes have at least one star?

1. First count *everything*: two symbols, 3 choices each, so by the multiplication principle $3 \times 3 = 9$ codes in all.

2. Now the opposite — codes with **no star**. Each symbol can only be moon or sun, 2 choices each: $2 \times 2 = 4$ codes.

3. Subtract: $9 - 4 =$ **5 codes** have at least one star.

4. (Check by listing the good ones: star-star, star-moon, star-sun, moon-star, sun-star — five, just as promised.)

WORKED EXAMPLE 2 — the opposite is “the special few”

How many of the numbers 1, 2, 3, …, 30 are NOT multiples of 4?

1. Everything first: there are 30 numbers in the list.

2. The opposite is the easy group — the multiples of 4: 4, 8, 12, 16, 20, 24, 28. That is 7 of them.

3. Subtract: $30 - 7 =$ **23 numbers** are not multiples of 4.

4. Counting the “not multiples” one by one would be a slog; the multiples are few and easy to list. Always count whichever side is smaller.

Watch how it works (one more)

WORKED EXAMPLE 3 — when you have to name the opposite yourself

Three friends each spin a spinner with 4 colors: red, green, blue, yellow. In how many ways do they NOT all land on the same color?

1. Everything first: three spins, 4 colors each, so $4 \times 4 \times 4 = 64$ results in all.

2. Here the opposite needs a moment of thought. The opposite of “not all the same” is “all the same” — all red, all green, all blue, or all yellow.

3. That opposite is tiny: just **4** results (one for each color).

4. Subtract: $64 - 4 =$ **60 ways**. The hard-sounding question becomes one subtraction once you spot what the opposite really is.

All three are the same move: count everything, count the opposite, subtract. The skill is spotting which group is the easy one.

Now you

Show the total and the opposite — not just the final number.

1. A 2-letter tag uses the letters X, Y, Z, and a letter may repeat. How many tags have at least one Z?

Everything (the total): ____________________

The ones I DON'T want: ____________________

Subtract: ________ – ________ = ________

2. How many of the numbers 1, 2, 3, …, 25 are NOT multiples of 5?

Everything (the total): ____________________

The ones I DON'T want: ____________________

Subtract: ________ – ________ = ________

3. A 3-digit code uses the digits 1, 2, 3, 4, and digits may repeat. How many codes have at least one 1?

Everything (the total): ____________________

The ones I DON'T want: ____________________

Subtract: ________ – ________ = ________

4. You flip a coin 4 times in a row, writing down heads or tails each time. In how many ways do you get at least one head?

Everything (the total): ________________

The ones I DON'T want: ________________

Subtract: ________ – ________ = ________

5. How many two-digit numbers have at least one digit that is a 3? (Remember a two-digit number can't start with 0.)

Everything (the total): ________________

The ones I DON'T want: ________________

Subtract: ________ – ________ = ________

6. Three friends each spin a spinner with 4 colors: red, green, blue, yellow. In how many ways do they NOT all land on the same color?

Everything (the total): ________________

The ones I DON'T want: ________________

Subtract: ________ – ________ = ________

Answer Key — for parents

What this packet teaches. One named technique — when what you want is hard to count head-on, count everything and subtract the part you don't want (Wanted = Total − Opposite) — shown first (the clubhouse hook, the bar picture, and three worked examples), then applied six times. It builds directly on Packet #3: the total and the opposite are each found with the multiplication principle. The three worked examples are deliberately different *kinds* of the same move: "at least one" where the opposite is "none" (1), "not a special kind" where the opposite is the easy handful (2), and "not all the same" where you must name the opposite yourself (3) — so every kind of problem below has been modeled before they meet it. Each problem has a three-row box printed under it: the goal is to see the total and the opposite, not just the total.

Answers.

1. Total 3 × 3 = 9 tags; opposite (no Z, so only X or Y) 2 × 2 = 4; so 9 − 4 = **5**. (Echoes Worked Example 1.)

2. Total 25 numbers; opposite (multiples of 5: 5, 10, 15, 20, 25) = 5; so 25 − 5 = **20**. (Echoes Worked Example 2.)

3. Total 4 × 4 × 4 = 64 codes; opposite (no 1, so 3 choices each) 3 × 3 × 3 = 27; so 64 − 27 = **37**.

4. Total 2 × 2 × 2 × 2 = 16 results; the opposite of "at least one head" is *no heads at all* — all four tails, just **1** result; so 16 − 1 = **15**. (The opposite can be a single case.)

5. Total two-digit numbers: 90 (10 through 99). Opposite (no 3 anywhere): the tens digit is 1–9 but not 3, so 8 choices, and the ones digit is 0–9 but not 3, so 9 choices — 8 × 9 = 72. So 90 − 72 = **18**. (The no-0-in-the-tens restriction is the trap, just like Packet #3's last problem.)

6. Total 4 × 4 × 4 = 64; opposite ("all the same": all red, all green, all blue, all yellow) = 4; so 64 − 4 = **60**. (Echoes Worked Example 3 — the opposite must be named.)

What to watch for. Three slips. First, **counting the hard side anyway** — if they start listing every code that has a star, steer them to the opposite ("which is easier to count, the ones with a star or the ones with none?"). Second, **forgetting to subtract** — they find the total and the opposite, then write one of those as the answer; the box's third row is there to force the subtraction. Third, **mis-naming the opposite** — on problem 6 the opposite of "not all the same" is "all the same" (not "all different"), and on problem 4 it is "all tails" (a single outcome). Problems 5 and 6 are the stretch: 5 hides the no-0 restriction inside the opposite count, and 6 makes them figure out what the opposite even is.

How to run it. Read the hook, the bar picture, and all three worked examples out loud together before they touch problem 1. The problems they do solo, in one sitting or two. This is a learning packet, not a test: helping on a stuck problem is fine, but help by pointing back at the matching worked example — and at the Total − Opposite box — not at the answer.